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First-Principles Study of CO, C₂H₂, and C₂H₄ Adsorption on Penta-Graphene for Transformer Oil Gas Sensing Applications

Min-Qi Zhu  and Xue-Feng Wang * 

Institute of Theoretical and Applied Physics, School of Physical Science and Technology, Soochow University,
1 Shizi Street, Suzhou 215006, China; zhuminqichn@163.com

* Correspondence: wxf@suda.edu.cn

Abstract

Penta-graphene, a novel two-dimensional carbon allotrope entirely composed of pentagonal carbon rings, has attracted increasing attention due to its unique geometric structure, mechanical robustness, and intrinsic semiconducting nature. In this study, we systematically investigate the adsorption behavior of three typical dissolved gases in transformer oil (CO, C₂H₂, and C₂H₄) on penta-graphene using first-principles calculations based on density functional theory. The optimized adsorption configuration, adsorption energy, charge transfer, adsorption distance, band structure, density of states, charge density difference, and desorption time are analyzed to evaluate the sensing capability of penta-graphene. Results reveal that penta-graphene exhibits moderate chemical interactions with CO and C₂H₂, accompanied by noticeable charge transfer and band structure changes, whereas C₂H₄ shows weaker physisorption characteristics. The projected density of states analysis further confirms the orbital hybridization between gas molecules and the substrate. Additionally, the desorption time calculations suggest that penta-graphene possesses good sensing and recovery potential, especially under elevated temperatures. These findings indicate that penta-graphene is a promising candidate for use in gas sensing applications related to the monitoring of dissolved gases in transformer oils.

Keywords: penta-graphene; oil dissolved gases; density functional theory; gas sensor



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1. Introduction

Two-dimensional (2D) materials have emerged as a focal point of research due to their remarkable physical, chemical, and electronic properties that differ significantly from their bulk counterparts [1–4]. Among these materials, graphene has long been celebrated for its extraordinary electrical conductivity, mechanical strength, and large surface area, making it a promising candidate for various applications including gas sensing [5–9]. However, the zero bandgap nature and chemical inertness of pristine graphene limit its practical utility in certain sensing applications where the selective and sensitive detection of gas molecules is required [10,11]. Recently, penta-graphene (PG), a novel 2D carbon allotrope composed entirely of pentagonal carbon rings, has been theoretically predicted and extensively studied as a material with distinctive structural and electronic properties [12,13]. Unlike graphene's hexagonal lattice, PG features a unique non-planar geometry with mixed *sp*²-*sp*³ hybridization, resulting in a wide bandgap approximately equal to that of gallium nitride and zinc oxide [14–16]. This sizable bandgap coupled with its superior mechanical properties and chemical reactivity makes PG a promising candidate for next-generation nanoelectronic and sensing devices [17–19].

In the context of power transformer diagnostics, the detection of fault characteristic gases such as carbon monoxide (CO), acetylene (C_2H_2), and ethylene (C_2H_4) is critical for early fault detection and condition monitoring [20–22]. These gases are generated due to thermal degradation, partial discharge, or arcing faults inside the transformer and serve as vital indicators of transformer health [23,24]. Conventional detection methods, including gas chromatography and infrared spectroscopy, although precise, suffer from drawbacks such as high cost, bulky instrumentation, and limited real-time monitoring capability [25–27]. Therefore, developing novel gas sensors that are sensitive, selective, and capable of rapid on-site detection is an urgent need.

While pristine graphene and other 2D materials have been widely studied for gas sensing, their interactions with small gas molecules are often weak, resulting in low sensitivity and selectivity. Modifications such as doping, defect engineering, or surface functionalization have been proposed to overcome these challenges [28–32]. Due to its unique lattice and mixed hybridization states, PG intrinsically exhibits enhanced chemical reactivity and potential for stronger interactions with gas molecules compared to graphene. These characteristics suggest that PG may provide improved adsorption and electronic response toward relevant gas molecules without the need for additional functionalization [33,34]. It is worth noting that penta-graphene is a theoretically predicted carbon allotrope and, to date, has not yet been synthesized experimentally. Nevertheless, its predicted properties continue to attract significant interest for potential applications in sensing and nanoelectronics.

Lima et al. have investigated the adsorption mechanism of oxygen molecules on defective PG and the results show a degree of selective sensing [35]. Lu et al. have studied the adsorption and separation behavior of CO_2 and N_2 on PG, and the results show that PG pore is a promising material for CO_2 capture and separation [36]. Li et al. explored the adsorption behavior of individual nucleobases as well as hydrogen-bonded base pairs from DNA and RNA on PG, revealing its potential as an effective platform for nanobiotechnological applications. Their findings indicate that PG holds promise for the selective detection of biomolecules, thereby highlighting its suitability in biosensing technologies [37]. Despite the promising attributes of PG, systematic studies focusing on its interaction with transformer fault gases are still scarce. Understanding the adsorption mechanisms, charge transfer behavior, and resultant electronic property changes upon gas molecule adsorption is essential for assessing the feasibility of PG as a gas sensing material.

In this work, we employ first-principles calculations based on density functional theory (DFT) to investigate the adsorption behavior and electronic properties of CO, C_2H_2 , and C_2H_4 molecules on PG. We analyze key parameters including adsorption energy (E_{ad}), charge transfer (Q_t), adsorption distance (d), band structure, density of states (DOS), charge density difference (CDD), and desorption time (τ) to elucidate the nature of the interactions and the sensitivity potential of PG toward these gases. The outcomes not only deepen the fundamental understanding of gas–surface interactions in PG systems but also provide theoretical insights toward the rational design of highly sensitive and selective gas sensors for transformer fault diagnostics and other environmental applications.

2. Computational Methods

First-principles calculations are carried out using the Quantum ATK software package [38,39]. The electronic structure calculations are carried out using the linear combination of atomic orbitals (LCAO) method, with the exchange–correlation interactions described by the Perdew–Burke–Ernzerhof (PBE) functional under the framework of the generalized gradient approximation (GGA) [40,41]. To account for long-range dispersion forces, the van der Waals interactions are treated using Grimme’s DFT-D3 dispersion correction scheme [42]. The interactions between valence electrons and nuclei are modeled using a

basic set of high PseudoDojo pseudopotentials [43,44]. A Monkhorst-Pack k-point mesh of $5 \times 5 \times 1$ and $9 \times 9 \times 1$ are utilized for the structural optimization and electronic structure calculations, respectively [45]. The density mesh cut-off energy is set to 150 Ha, and a vacuum layer of 20 Å is applied along the z axis to prevent image interactions. Periodic boundary conditions are employed in the x axis and y axis. The high-symmetry path Γ -X-S-Y- Γ is chosen for the calculations. The geometry optimization is performed using the limited-memory Broyden–Fletcher–Goldfarb–Shanno (LBFGS) algorithm, and convergence is achieved when the maximum atomic force is less than 0.01 eV/Å [46]. The self-consistent field (SCF) convergence is achieved with an energy threshold of 10^{-5} eV.

3. Results and Discussions

Figure 1 illustrates the structural and electronic properties of pristine PG. As shown in Figure 1a, the optimized structure of PG consists entirely of pentagonal carbon rings arranged in a non-planar configuration. The structure possesses $P\bar{4}2_1m$ symmetry (space group No. 113) with a tetragonal lattice. The optimized lattice constant is $a = b = 3.64$ Å, and the C–C bond lengths are found to be 1.34 Å and 1.55 Å, respectively. These structural parameters are in good agreement with previous reports [13,34,47]. For subsequent calculations involving gas adsorption, we constructed a 3×3 supercell, which contains 54 carbon atoms, to ensure minimal interaction between the periodic images of the adsorbed molecules. The band structure of pristine PG is shown in Figure 1b. The material exhibits an indirect bandgap of 2.28 eV, with the valence band maximum (VBM) located at the X point and the conduction band minimum (CBM) at the S point, which is in agreement with previously calculated values [35,37,48]. This semiconducting nature is further supported by the total density of states (TDOS) illustrated in Figure 1c, which clearly shows a bandgap around the Fermi level. These findings demonstrate that PG possesses favorable structural stability and an intrinsic bandgap, which are essential prerequisites for gas sensing applications.

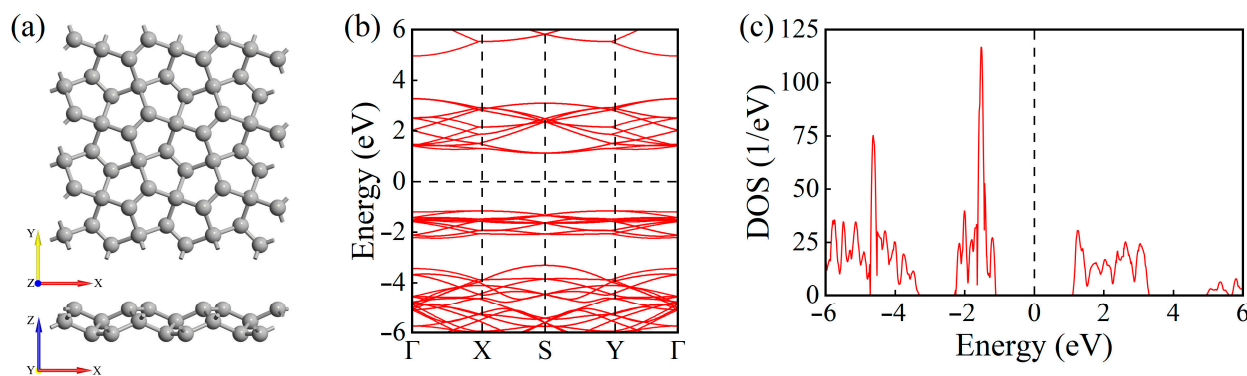


Figure 1. (a) Top and side views of the optimized structure of PG. (b) Band structure and (c) DOS of PG. A black dashed line denotes the Fermi energy, which is set to 0 eV.

Before analyzing the adsorption behavior on the penta-graphene surface, it is necessary to investigate the geometric configurations of the isolated gas molecules. As shown in Figure 2a, the CO molecule adopts a linear configuration with a bond length of 1.14 Å between the carbon and oxygen atoms. For C₂H₂, as shown in Figure 2b, the molecule exhibits a linear structure with C–H and C–C bond lengths of 1.08 Å and 1.21 Å, respectively. In contrast, the C₂H₄ shown in Figure 2c possesses a planar geometry due to the sp^2 hybridization of carbon atoms, with C–C and C–H bond lengths of 1.34 Å and 1.09 Å, respectively. The corresponding bond angles are 116.64° and 121.57°, consistent with the expected trigonal planar arrangement. All optimized geometries are consistent with

experimental and theoretical results, and thus we will use this as a starting point for the construction of the adsorption system [49–51].

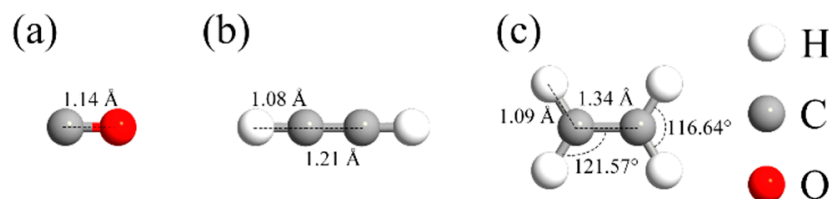


Figure 2. The optimized structure of (a) CO, (b) C₂H₂, and (c) C₂H₄ gas molecules. H, C, and O atoms are shown by white, gray, and red balls, respectively.

To explore the sensing ability of PG toward transformer fault gases, we investigate the adsorption behavior of CO, C₂H₂, and C₂H₄ molecules on the PG surface. For each gas species, various initial configurations are considered, including orientations parallel and perpendicular to the surface, with multiple adsorption sites tested. The initial adsorption distance is set to 3.00 Å for all molecules prior to geometry optimization. After full structural relaxation, the most stable configurations are presented in Figure 3. As shown in Figure 3a, the CO molecule prefers a perpendicular orientation, with the C atom binding directly atop the surface C atoms, and the O atom pointing away from the surface. The measured distance of the C atom from the surface is 1.26 Å, indicating a strong interaction. In Figure 3b, the C₂H₂ molecule adopts a parallel adsorption configuration and binds to the PG surface through its carbon atoms at a distance of 1.51 Å. In contrast, as shown in Figure 3c, the C₂H₄ molecule remains nearly parallel to the surface without forming any chemical bonds, maintaining a relatively long adsorption distance of 2.97 Å, which suggests weak interaction. These optimized structures form the basis for the further analysis of adsorption energy, charge transfer, and electronic properties discussed in the following sections. The relevant parameters are shown in Table 1.

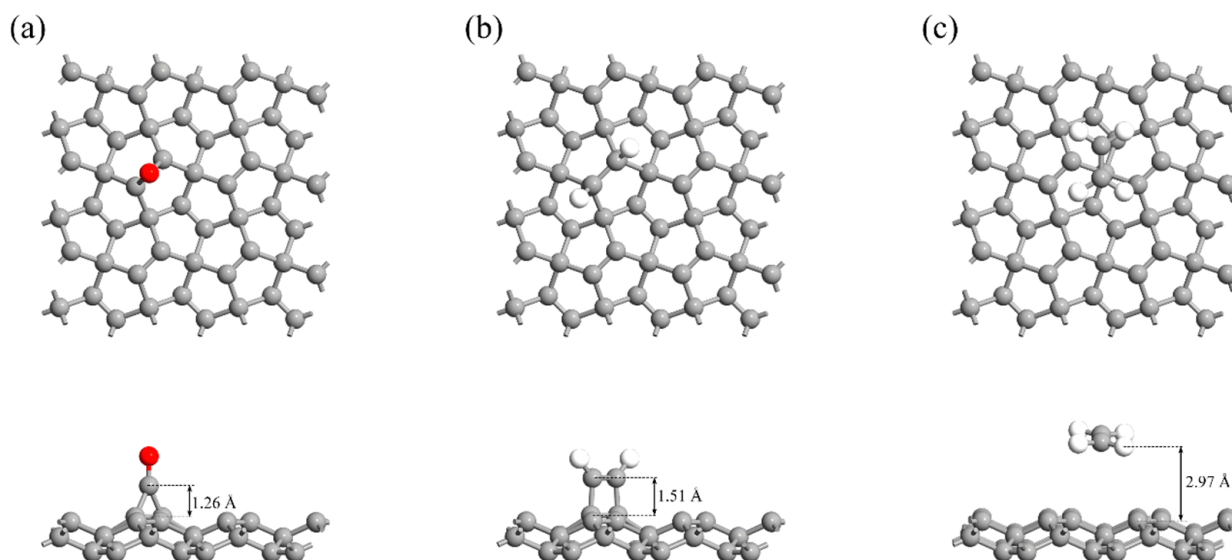


Figure 3. Top and side views of the most stable configuration for the adsorption of (a) CO, (b) C₂H₂, and (c) C₂H₄ on PG.

Table 1. The adsorption distance (d), adsorption energy (E_{ad}), bandgap (E_g), charge transfer (Q_t), and desorption time (τ) of three characteristic gases adsorbed on PG.

System	d (Å)	E_{ad} (eV)	E_g (eV)	Q_t (e)	τ (s)
PG/CO	1.26	−1.32	1.80	0.18	5.12×10^4 (398 K)
					6.99×10^2 (448 K)
					2.26×10^1 (498 K)
PG/C ₂ H ₂	1.51	−2.99	1.91	−0.23	7.08×10^{25} (398 K)
					4.22×10^{21} (448 K)
					1.78×10^{18} (498 K)
PG/C ₂ H ₄	2.97	−0.09	2.23	−0.02	1.38×10^{-11} (398 K)
					1.03×10^{-11} (448 K)
					8.14×10^{-12} (498 K)

The adsorption energy is calculated using the formula:

$$E_{ad} = E_{PG/gas} - E_{PG} - E_{gas} \quad (1)$$

where $E_{PG/gas}$, E_{PG} , and E_{gas} represent the total energies of the adsorbed system, pristine PG, and isolated gas molecule, respectively. The computed E_{ad} for CO, C₂H₂, and C₂H₄ are −1.32 eV, −2.99 eV, and −0.09 eV, respectively. Therefore, combined with the adsorption distance, C₂H₂ shows strong chemisorption, CO exhibits moderate chemisorption, and C₂H₄ is weakly physisorbed on the PG surface. These results are consistent with the charge transfer and electronic structure analyses presented in the following sections, confirming the stronger interaction between PG and CO or C₂H₂, which are more likely to induce significant changes in the material's electronic properties and are thus more suitable for gas sensing.

To investigate the influence of gas adsorption on the electronic properties of PG, we calculate the band structures of the system after the adsorption of CO, C₂H₂, and C₂H₄ molecules, as shown in Figure 4. As shown in Figure 4a, upon CO adsorption, new impurity states emerge near the Fermi level, and the bandgap is significantly reduced to 1.80 eV compared to the pristine PG. This indicates the strong interaction and charge transfer between the CO molecule and the substrate, which could result in notable changes in the electrical conductivity, which is a basic criterion for effective gas sensing. As shown in Figure 4b, in the case of the adsorbed C₂H₂ molecule, the band structure also exhibits notable modifications. A shallow impurity state appears slightly above the VBM with a measured bandgap of 1.91 eV. These changes suggest moderate coupling between C₂H₂ and the substrate, which may also induce detectable variations in conductivity. Conversely, for the adsorption of C₂H₄ molecules, as shown in Figure 4c, the band structure remains nearly unchanged, preserving a wide bandgap of 2.23 eV similar to the pristine case. This implies a weak physisorption with negligible electronic perturbation, which is consistent with the larger adsorption distance and smaller adsorption energy. These results demonstrate that PG exhibits molecule-dependent electronic responses, with stronger interactions and detectable band modifications for CO and C₂H₂, indicating its potential as a selective sensor material for transformer oil dissolved gases.

To further understand the electronic interactions between PG and the gas molecules, the total density of states (TDOS) before and after the adsorption of CO, C₂H₂, and C₂H₄ are calculated, as shown in Figure 5. In each case, the TDOS of pristine PG is displayed in black, while the red curve represents the corresponding gas-adsorbed system. For the CO-adsorbed system, as shown in Figure 5a, significant changes are observed near the Fermi level. New electronic states appear within the bandgap region, indicating strong

orbital hybridization and electronic coupling between CO and the PG surface. These changes are consistent with the reduced bandgap observed in the band structure analysis, and suggest an increase in carrier density, which is favorable for gas sensing. In the case of C_2H_2 adsorption, as shown in Figure 5b, the TDOS also exhibits noticeable modifications, with shallow defect states emerging near the valence band edge. These features imply a moderate interaction strength and potential variations in electrical conductivity upon gas exposure. In contrast, the TDOS of the C_2H_4 -adsorbed system, as shown in Figure 5c, shows minimal deviation from the pristine case. No prominent states appear near the Fermi level, suggesting weak physisorption and limited charge transfer between C_2H_4 and the substrate. This weak electronic interaction correlates with the minimal structural and band structure perturbation observed earlier. Overall, the TDOS results support the conclusion that PG exhibits molecule-specific electronic responses, particularly showing higher sensitivity toward CO and C_2H_2 , which makes it a promising material for the selective detection of gases dissolved in transformer oil.

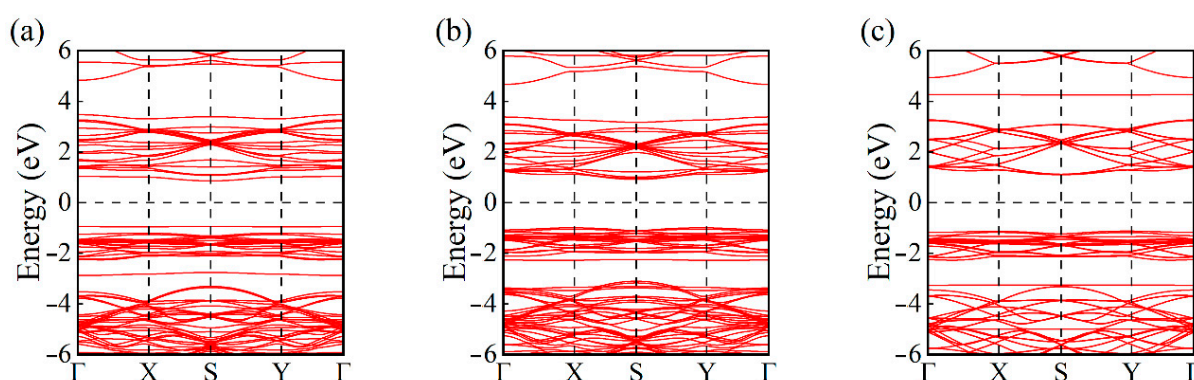


Figure 4. Band structures of the PG adsorption system for (a) CO, (b) C_2H_2 , and (c) C_2H_4 . A black dashed line denotes the Fermi energy, which is set to 0 eV.

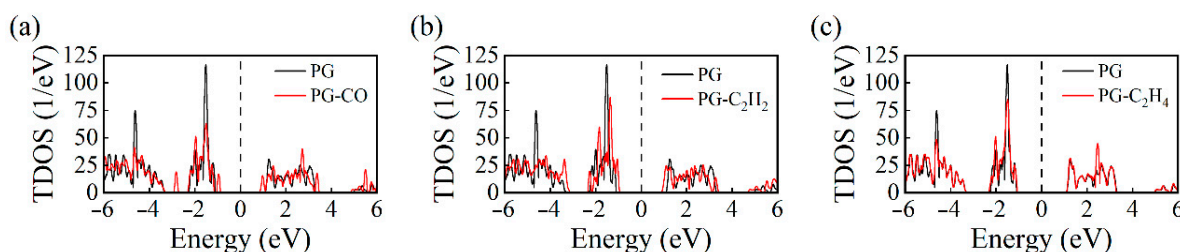


Figure 5. TDOS of PG adsorption system for (a) CO, (b) C_2H_2 , and (c) C_2H_4 . A black dashed line denotes the Fermi energy, which is set to 0 eV.

To gain deeper insight into the electronic interactions and orbital hybridization between the adsorbed gas molecules and PG, the projected density of states (PDOS) is calculated for each gas-adsorbed system, as shown in Figure 6. The PDOS reflects the contribution of specific atomic orbitals to the total density of states by projecting the Kohn–Sham eigenstates onto selected orbitals [52]. For the CO-adsorbed system, as shown in Figure 6a, a significant overlap between the p -orbitals of O and the p -orbitals of C atoms is observed near the Fermi level, suggesting strong orbital hybridization and charge delocalization. These features indicate that chemisorption has occurred, leading to the formation of new electronic states and enhanced electrical conductivity, both of which are beneficial for gas sensing. In the case of C_2H_2 adsorption, as shown in Figure 6b, the new electronic states near the Fermi energy level are mainly contributed by the p -orbitals of C atoms, which can be attributed to the C atoms in C_2H_2 and PG. In contrast, only a slight orbital overlap between the s -orbitals of H atoms and p -orbitals of C atoms is observed near -3.30 eV. This

suggests partial charge transfer and a degree of electronic coupling, which aligns with the earlier TDOS and band structure observations. For C_2H_4 adsorption, as shown in Figure 6c, only a slight contribution from the s -orbitals of H atoms is observed near -5.00 eV, which corresponds to the emerging states in the TDOS of Figure 5c, whereas the states at the other energies remains almost unchanged. This suggests that the interaction between C_2H_4 and PG is weak and mainly governed by van der Waals forces, thus confirming its physisorption nature. These PDOS results further support the conclusion that PG exhibits stronger electronic interactions with CO and C_2H_2 compared to C_2H_4 , reinforcing its potential as a selective sensing material for transformer oil gases.

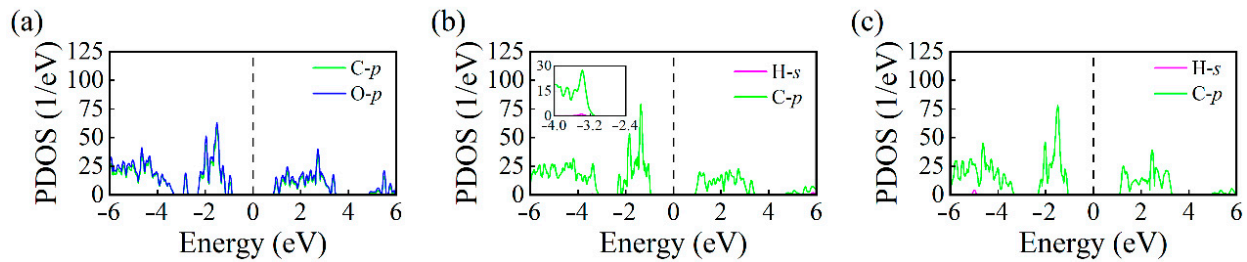


Figure 6. PDOS of PG adsorption system for (a) CO, (b) C_2H_2 , and (c) C_2H_4 . A black dashed line denotes the Fermi energy, which is set to 0 eV.

To elucidate the electronic interactions between gas molecules and the PG surface, charge density difference calculations are performed for the CO, C_2H_2 , and C_2H_4 adsorption systems. The CDD is defined as

$$\Delta\rho = \rho_{PG/gas} - \rho_{PG} - \rho_{gas} \quad (2)$$

where $\rho_{PG/gas}$, ρ_{PG} , and ρ_{gas} denote the total charge density of the adsorbed system, pristine PG, and isolated gas molecule, respectively. In addition, the charge transfer during the adsorption process based on the Mulliken population method is calculated by the following equation:

$$Q_t = Q_{after} - Q_{before} \quad (3)$$

where Q_{after} and Q_{before} are the total charges of the gas molecule after and before adsorption, respectively. As illustrated in Figure 7, the CDD isosurfaces clearly exhibit regions of charge exhaustion (magenta) and accumulation (cyan), providing insight into the nature of gas–substrate interactions. For CO and C_2H_2 adsorption, as shown in Figure 7a,b, apparent charge redistribution occurs at the interface, indicating strong coupling between the gas and substrate. This observation suggests the presence of chemisorption, likely driven by orbital hybridization and charge transfer mechanisms. Quantitatively, as summarized in Table 1, the CO gains 0.18 e from PG, while C_2H_2 and C_2H_4 donate 0.23 e and 0.02 e, respectively. These values further confirm the stronger coupling of CO and C_2H_2 with PG. In contrast, for the C_2H_4 adsorption system, as shown in Figure 7c, a significantly lower isosurface value of $0.0005 \text{ e} \cdot \text{\AA}^{-3}$ is applied to visualize the subtle electron redistribution. The charge transfer between C_2H_4 and PG is minimal, and the interaction is predominantly governed by weak van der Waals forces, characteristic of physisorption. Overall, these results demonstrate that CO and C_2H_2 exhibit stronger electronic interactions with PG, which is conducive to enhanced sensing response. Conversely, the weak interaction of C_2H_4 suggests limited sensitivity and selectivity toward this molecule.

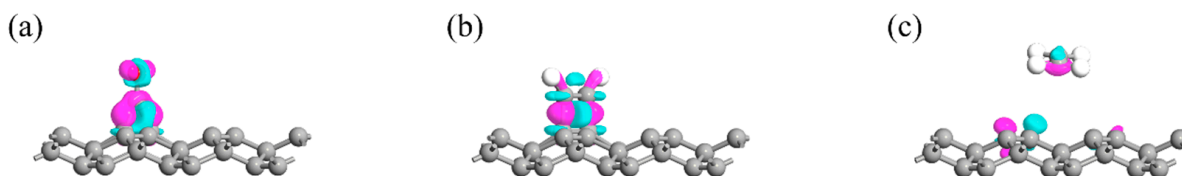


Figure 7. The CDD of the PG adsorption system for (a) CO, (b) C₂H₂, and (c) C₂H₄. The magenta and cyan regions represent electron exhaustion and accumulation, respectively. The isosurface is 0.01 e·Å^{−3} for (a) and (b), and 0.0005 e·Å^{−3} for (c).

The desorption time of the gas molecules from the sensing surfaces is a key parameter for evaluating the recovery performance and reusability of gas sensors. In this study, based on transition state theory and the Van't-Hoff–Arrhenius expression, the desorption time of CO, C₂H₂, and C₂H₄ molecules on PG is calculated using the following equation [53]:

$$\tau = \nu_0^{-1} \exp(-E_{ad}/k_B T) \quad (4)$$

where ν_0 represents the trial frequency ($1 \times 10^{12} \text{ s}^{-1}$), k_B is the Boltzmann constant ($8.62 \times 10^{-5} \text{ eV/K}$), and T is the temperature (K). Figure 8 shows the variation in desorption time at temperatures of 398 K, 448 K, and 498 K, which are relevant to the operating conditions of power transformers. The results indicate that the desorption time of C₂H₄ remains relatively short across the studied temperature range, implying weak adsorption and fast recovery, characteristic of physisorption. In contrast, CO and C₂H₂ exhibit significantly longer desorption times at lower temperatures, especially for C₂H₂, suggesting stronger interactions with the PG surface and slower desorption kinetics. However, as the temperature increases, the desorption times of all three gases decrease exponentially, confirming the thermally activated nature of the desorption process. These findings show that PG can achieve controlled desorption behavior for gas adsorption, especially at high temperatures. The desorption time is very favorable for the practical applications of gas sensors where fast response and reset cycles are essential.

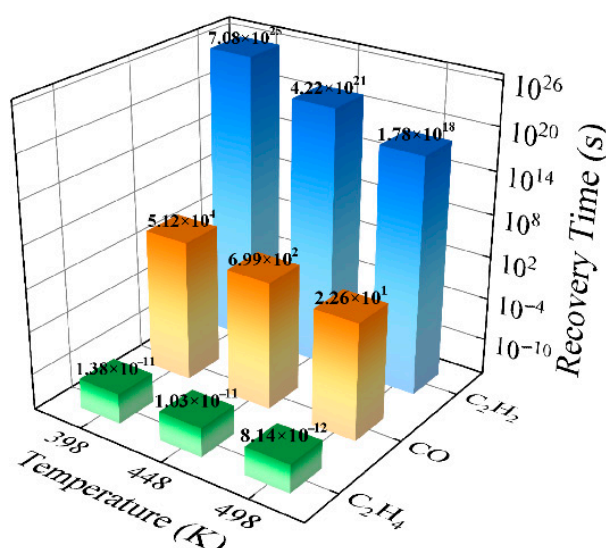


Figure 8. Desorption time of the PG adsorption system at three different temperatures.

4. Conclusions

In this work, we performed a comprehensive DFT-based study on the adsorption properties of CO, C₂H₂, and C₂H₄ molecules on PG, aiming to evaluate its potential as a

sensing material for transformer oil-dissolved gases. The key findings are summarized as follows:

1. Geometric and electronic properties: PG exhibits an indirect bandgap of 2.28 eV with a non-planar lattice structure. Its unique sp^2 - sp^3 hybridized geometry provides reactive sites for molecular adsorption.
2. Adsorption behavior: CO and C₂H₂ display moderate chemisorption on PG with adsorption energies of −1.32 eV and −2.99 eV, respectively, along with significant charge transfer and local electronic structure modifications. C₂H₄ shows a weaker adsorption energy of −0.09 eV, indicating a typical physisorption behavior.
3. Electronic Structure Analysis: Both TDOS and PDOS reveal that CO and C₂H₂ adsorption induce notable changes near the Fermi level, suggesting an enhanced sensing response. CDD plots support the presence of electron redistribution upon adsorption.
4. Desorption dynamics: Calculated desorption times indicate that CO and C₂H₂ molecules are more strongly bound to the surface, requiring thermal activation for rapid desorption, whereas C₂H₄ can be readily desorbed at ambient temperatures.

Overall, PG exhibits selective and thermally tunable adsorption behavior toward CO and C₂H₂, making it a viable material for gas sensor devices aimed at monitoring transformer oil decomposition products.

Author Contributions: Conceptualization, methodology, software, M.-Q.Z. and X.-F.W.; formal analysis, investigation, data curation, writing—original draft preparation, visualization, M.-Q.Z.; validation, resources, writing—review and editing, supervision, project administration, funding acquisition, X.-F.W. All authors have read and agreed to the published version of the manuscript.

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